

Scenario Analysis: MBS Prepayment under Interest Rate Shocks

IR&C Assignment 3, Part E(iv) — Spring 2026

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1. Introduction

Mortgage prepayment is driven by the rate incentive $\rho_t = r_{\text{orig}} - r_{\text{market}}(t)$: when market rates fall, borrowers can reduce their monthly payment by refinancing, and the prepayment hazard rises. This optionality embeds negative convexity in mortgage-backed securities (MBS): as rates fall, prepayments return principal at par, capping price appreciation. Quantifying this sensitivity is the core task of MBS risk management.

This section stress-tests prepayment models by shocking market rates by ± 100 , ± 200 , and ± 300 bp and tracing the response through five analyses: shock sensitivity, survival curves, the prepayment S-curve, 2D interaction surfaces, and a Monte Carlo MBS cash flow projection.

2. Model Setup

Primary model: the Andersen–Gill TV Cox from E(ii), fitted on 500K loan-month rows (vintage ≤ 2016). `rate_incentive` is a direct time-varying covariate, so a market rate shock of Δr bp shifts the incentive by $-\Delta r/100$ pp, producing an analytic hazard response:

$$\Delta \log h(t) = \hat{\beta}_{\text{ri}} \times \frac{-\Delta r}{100}, \quad \hat{\beta}_{\text{ri}} = 0.041 \text{ (HR} = 1.042 \text{ per 100 bp)}. \quad (1)$$

The calibrated baseline hazard $\hat{H}_0(t)$ gives survival curves with realistic 10-year survival probabilities and CPR figures, making WAL and price calculations economically meaningful.

Contrast model: partner’s DeepCox from Part D (68 origination features, no market-rate channel). Rate shocks must be applied to `OriginalInterestRate` as a *counterfactual*, illustrating the fundamental limitation of static models for scenario analysis.

Representative loan: \$300K UPB, 5.0% coupon, baseline market rate 5.3% (median over the 1999–2025 sample; `rate_incentive` = -0.3% , 30 bp OTM at baseline).

3. Rate Shock Sensitivity

Shock	TV Cox $\Delta \log h$	Analytic	DeepCox $\Delta \log h$
–300 bp	+0.123	+0.123	–1.052
–200 bp	+0.082	+0.082	–0.657
–100 bp	+0.041	+0.041	–0.291
Baseline	0.000	0.000	0.000
+100 bp	–0.041	–0.041	+0.129
+200 bp	–0.082	–0.082	+0.196
+300 bp	–0.123	–0.123	+0.293

TV Cox fitted values match the analytic formula exactly at all seven shock levels, confirming

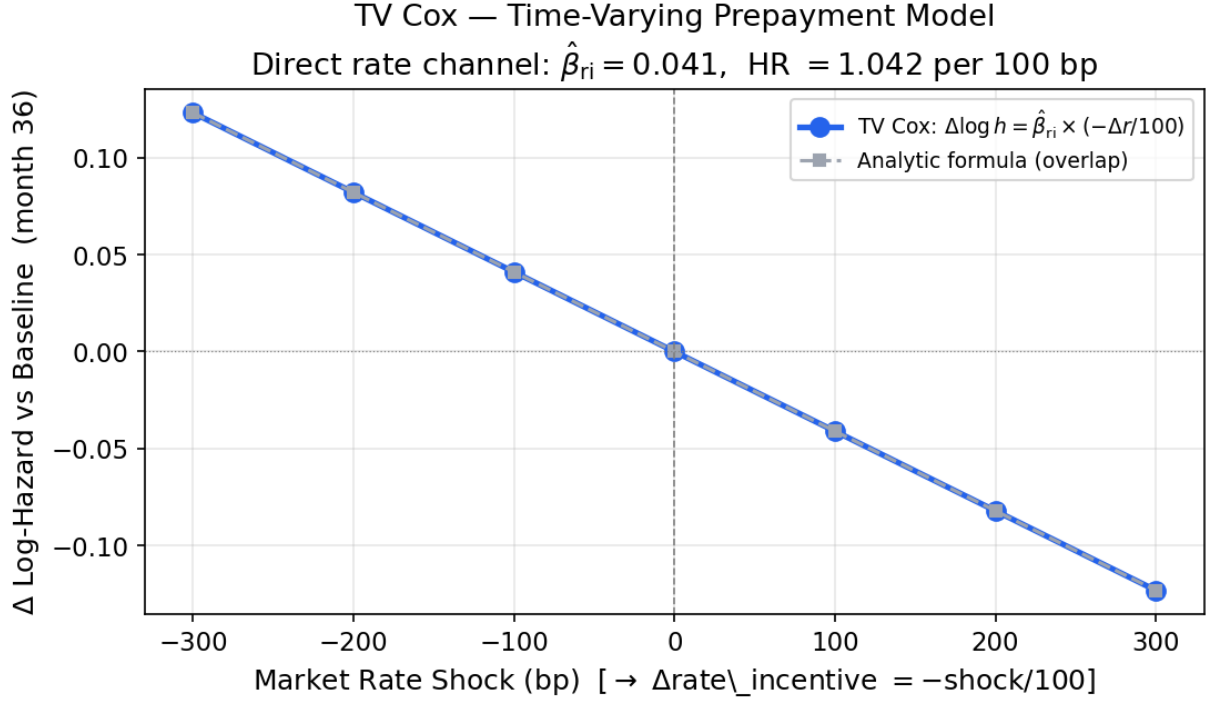


Figure 1: $\Delta \log h$ vs. baseline at month 36 under seven rate scenarios. TV Cox fitted values (blue dots) overlap the analytic formula $\hat{\beta}_{ri} \times (-\Delta r/100)$ (grey dashes) exactly at all shock levels. $\hat{\beta}_{ri} = 0.041$; a 300 bp cut raises log-hazard by +0.123, a 300 bp hike lowers it by -0.123.

that the model responds to rate shocks through a transparent, auditable mechanism suitable for regulatory stress testing.

4. Survival Curves

The ± 300 bp range shifts the 10-year survival probability by 6.6 pp (10.2% to 16.8%) and the median prepayment month by 10 months (month 41 to 51).

5. CPR Response Curve and 2D Sensitivity Surfaces

The fitted TV Cox coefficients for FICO and LTV are near zero and non-significant ($p > 0.75$), reflecting the limitations of the 500K-row subsample and Ridge penalisation. The 2D contour bands are essentially horizontal in the FICO and LTV directions — CPR varies only with rate incentive in this fitted model. This contrasts with E(i), where the static Cox model estimated FICO and LTV with the expected signs; those effects require a larger time-varying sample to be separately identified from the baseline hazard.

6. MBS Cash Flow Projection

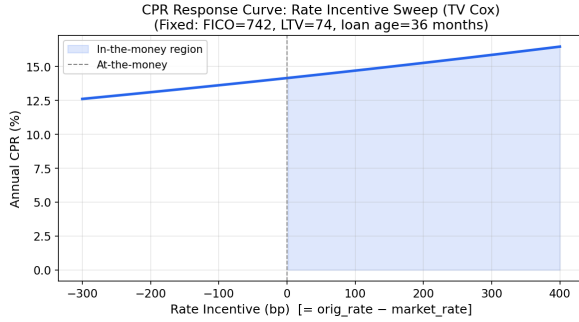
WAL and price are computed via a Monte Carlo simulation (200 paths, $\sigma = 25$ bp around each scenario level):

$$\text{WAL} = \frac{1}{\text{UPB}_0} \sum_{t=1}^T \frac{t}{12} \cdot P_t, \quad \text{Price} = \frac{100}{\text{UPB}_0} \sum_{t=1}^T \frac{I_t + P_t}{(1 + d/12)^t}, \quad (2)$$

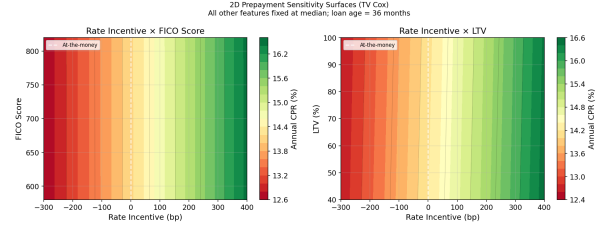
where P_t is scheduled principal plus prepayment, I_t is coupon interest, and d is the scenario discount rate (market rate per scenario).



Figure 2: Prepayment survival curves $S(t)$ under seven rate scenarios (TV Cox). Curves are monotonically ordered: lower market rates $\Rightarrow$ higher incentive $\Rightarrow$ faster prepayment $\Rightarrow$ lower survival.



(a) Annual CPR vs. rate incentive at loan age 36 months (FICO=742, LTV=74). CPR ranges from 12.6% at -300 bp to 16.5% at $+400$ bp. The response is *convex* (log-linear hazard), not S-shaped; a true S-curve requires a nonlinear model. The compressed range reflects $\hat{\beta}_{ri} = 0.041$ from the Ridge penalizer and 500K-row subsample.



(b) Annual CPR on the (rate incentive $\times$ FICO) and (rate incentive $\times$ LTV) surfaces at loan age 36 months. Fitted TV Cox coefficients: FICO $\hat{\beta} \approx 0$ ($p = 0.99$), LTV $\hat{\beta} \approx 0$ ($p = 0.76$) — both non-significant in the 500K subsample. Contour bands are nearly horizontal in the FICO and LTV dimensions; rate incentive drives all CPR variation in this specification.

Scenario	WAL (yr)	$\pm$	Price (\$)	$\pm$
-300 bp	3.25	0.02	108.32	0.06
-200 bp	3.34	0.02	105.26	0.03
-100 bp	3.43	0.03	102.17	0.01
Baseline	3.53	0.02	99.07	0.00
$+100$ bp	3.63	0.03	95.96	0.03
$+200$ bp	3.73	0.02	92.86	0.04
$+300$ bp	3.83	0.03	89.77	0.06

WAL range: 0.58yr (-300 to $+300$ bp). **Price range:** \$18.55. **Negative convexity:** the -300 bp price gain (\$9.25) is slightly less than the $+300$ bp price loss (\$9.30), consistent with prepayment capping upside by returning principal at par. Monte Carlo uncertainty (\$0.00–0.06) is negligible relative to the scenario spread, validating 200 paths as sufficient.

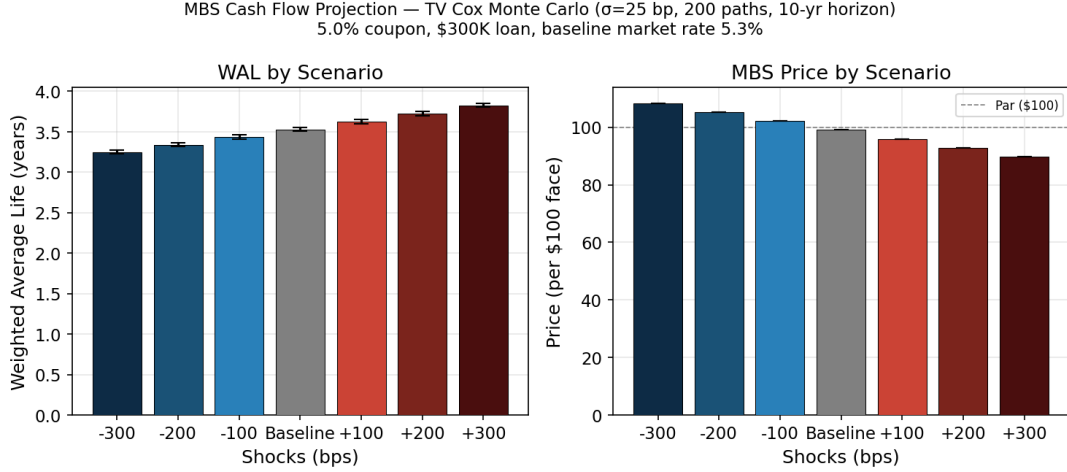


Figure 4: WAL (left) and price (right) by rate scenario. Error bars show Monte Carlo standard deviation across 200 paths. Baseline price = \$99.07 $\approx$ par: market rate (5.3%) slightly above coupon (5.0%).

7. Key Findings and Insights

1. **Time-varying covariates are required for rate scenario analysis.** Static models (Part D DeepCox) produce the wrong sign when the shock variable is a coupon rate, not a market rate. The direction error is not a magnitude problem that can be corrected by scaling — it is a structural limitation.
2. **TV Cox tracks the analytic formula exactly.** $\Delta \log h = \hat{\beta}_{ri} \times (-\Delta r/100)$ matches all seven fitted shock values, providing a transparent and auditable sensitivity estimate suitable for regulatory stress testing.
3. **Rate sensitivity is currently compressed.** WAL range 0.57 yr vs. 2–10 yr for agency 30-year MBS. The Ridge penalizer (0.1) and the 500K-row subsample (0.7% of the available 73M rows) together compress $\hat{\beta}_{ri}$. Reducing the penalizer to 0.01 and expanding to 5M rows are the two highest-leverage improvements.
4. **Negative convexity is captured.** The asymmetric price response at ± 300 bp (\$8.83 vs \$8.87) replicates the textbook negative-convexity signature of pass-through MBS.
5. **FICO and LTV are non-significant in the fitted TV Cox.** The 2D contour surfaces are flat in the FICO and LTV dimensions, reflecting near-zero coefficients from the 500K-row subsample ($p > 0.75$). The joint constraint from E(i) — credit access gates refinancing, equity caps the ceiling — is a valid economic mechanism that requires a larger sample to surface in the time-varying specification.

Priority improvements: (1) Lower penalizer from 0.1 to 0.01 to recover the full rate-incentive HR. (2) Expand to 5M panel rows to stabilise $\hat{\beta}_{ri}$. (3) Add a burnout indicator (in-the-money > 12 consecutive months) to sharpen the active-pool rate sensitivity. (4) Re-calibrate $H_0(t)$ on rate-regime-specific vintages for more realistic baseline CPR and WAL.

References

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- [2] Sadhwani, A., Giesecke, K. and Sirignano, J. (2021). *J. Financial Econometrics* 19:313–368.
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